Jonas Rieger

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https://jonasrieger.github.io/

Education

Doctoral degree *[Doktor der Naturwissenschaft]*, Dr. rer. nat. | ***Statistics*** — Sept. 2022
TU Dortmund University — Dortmund, Germany

Master of Science, M. Sc. | ***Statistics*** — Nov. 2018
TU Dortmund University — Dortmund, Germany

Bachelor of Science, B. Sc. | ***Statistics*** — Oct. 2016
TU Dortmund University — Dortmund, Germany

Previous and Current Positions

Scientific Manager of TRR 391 — since Oct. 2024
TU Dortmund University — Dortmund, Germany

Postdoc — since Oct. 2022
TU Dortmund University — Dortmund, Germany

NLP Scientist — Oct. 2022 – Sept. 2023
Leibniz Institute for Media Research | Hans-Bredow-Institut (HBI) — Hamburg, Germany

Doctoral student — Dec. 2018 – Sept. 2022
TU Dortmund University — Dortmund, Germany

Affiliations

Member of the TU Dortmund Young Academy — since June 2023
Member of DoCMA (Dortmund Center for Data-based Media Analysis) — since Dec. 2018
Leibniz Institute for Media Research | Hans-Bredow-Institut (HBI) — Oct. 2023 – Sept. 2024

Funding

BMWK: Federal Ministry for Economic Affairs and Climate Action — Sept. 2024 – Feb. 2027
Joint project: Social (in)justice in the energy transition — From the digital debate to the living world — 100 875€
Sub-project: Monitoring narratives about the energy transition

RC Trust: Research Center for Trustworthy Data Science and Security — Oct. 2023
Trustworthy performance evaluation of large language models — 21 000€

TU Dortmund Young Academy — June 2023
The era of ChatGPT: Evaluation and regulation of large language models — 5000€

Professional Experience

Preparation and submission of an approved CRC/Transregio grant application — Oct. 2024 – June 2028
TRR 391: Spatio-temporal statistics for the transition of energy and transport — DFG

Projects

GADMO — since Feb. 2025; Oct. 2022 – Sept. 2023
German-Austrian Digital Media Observatory — EU

Diskurs Energiewende — since Sept. 2024
Monitoring narratives about the energy transition — BMWK

NEAR — since Oct. 2023
Narrative Economics Alliance Ruhr — MERCUR

Further Project Affiliations

Medien-Doktor Assistance since Mar. 2023
KI-Assistenzsysteme für eine bessere Medizinberichterstattung

GADMO since Oct. 2022
German-Austrian Digital Media Observatory EU

FLACA Oct. 2022 – Sept. 2024
Few-Shot Learning for Automated Content Analysis in Communication Science BMBF

Research Interests

Methodological research

- Evaluation of NLP systems (e.g., language models, topic models) in terms of quality, reliability, robustness
- Parameter-efficient fine-tuning (PEFT) for large language models in few-shot scenarios
- Annotator disagreement and its incorporation into the modeling
- Model selection and parameter tuning of topic models
- Update algorithms and monitoring settings for topic models
- Detection of structural breaks, events and narratives in text corpora

Software engineering in R, Python, and Shiny

- Author and maintainer of CRAN packages rollingIda, ldaPrototype, and topiclabels
- Author and maintainer of Python module petapter
- Co-author and contributor of CRAN packages tosca and spINAR
- Co-author and co-project leader of the Medien-Doktor Assistance App, a Shiny app to assist in evaluating incoming and outgoing media articles, currently used by editorial teams at tagesschau, NDR Info and Nürnberger Nachrichten

Applied research

- Text corpus-based indicators, e.g., UPI and IPI
- Content analysis of texts and tweets of political parties and parliamentarians
- Argument mining and extracting narrative elements from news and social media debates
- Characteristics of disinformation and fact-checks

Service to the Research Community

Reviewing

- ACL Rolling Review and ACL-associated workshops (18 papers)
- Advances in Statistical Analysis
- Applied Sciences
- Communication Methods and Measures
- Comparative European Politics
- Computational Intelligence
- Educational and Psychological Measurement
- Electronics
- Frontiers in Artificial Intelligence
- Frontiers in Social Psychology
- PLOS One
- Scientometrics
- Statistical Papers: DOI: 10.1007/s00362-019-01126-7
- Sustainability

Conferences

- Program committee of workshop ClimateNLP at EMNLP 2026
- Organizing committee of NEAR Conference 2024
- Program committee of workshop ClimateNLP at ACL 2024
- Organizing committee of Statistische Woche 2023

Academic Administration

- Department Board (Fakultätsrat) since July 2024

Invited Talks

Research Colloquium at Universidad Católica del Norte, UCN — Nov. 2025, Antofagasta, Chile
Narrative(s of) transition - Monitoring the German "Energiewende" across media channels

Research Colloquium of the Faculty of Business Studies and Economics, JLU — Jan. 2025, Gießen, Germany
Detecting narration changes in economics
utilizing continuous topic modeling and (large) language models

CMStatistics 2024 — Dec. 2024, London, England
Monitoring (social) media narratives
combining retrospective few-shot classification with continuous topic modeling

CompStat 2024 — Aug. 2024, Gießen, Germany
PETapter: A masked-language-modeling classification head
for modular fine-tuning of (large) language models

BDI/BDA-Arbeitskreis Statistik — May 2024, Düsseldorf, Germany
Bekämpfung von Desinformation und Fake-News durch GADMO

ZPID Lecture Series — Dec. 2022, Trier, Germany
Keep rollin'! The abilities for monitoring growing corpora using RollingLDA

Contributions to Conferences and Workshops

Text2Story'26 Workshop @ECIR 2026 — Mar. 2026, Delft, The Netherlands
Following socio-environmental conflict narratives about energy transition in Chile:
A spatio-temporal analysis using dynamic topic modeling

BEWEGT 2026 — Mar. 2026, Berlin, Germany
Von Worten zum Monitoring: NLP-Tools für die automatische Textanalyse
und Visualisierung gesellschaftlicher Diskurse zur Energiewende

KONVENS 2025 — Sept. 2025, Hildesheim, Germany
PETapter: Leveraging PET-style classification heads
for modular few-shot parameter-efficient fine-tuning

DigiKomm & Methoden 2024 — Sept. 2024, Hamburg, Germany
Classifying the needle in the haystack? Problemstellungen beim Einsatz von Argument-Mining für kommunikationswissenschaftliche Fragestellungen am Beispiel der Waffenlieferungsdebatte

DGPuK 2024 — Mar. 2024, Erfurt, Germany
Exploring the potential of large language models (such as GPT-4) for
(semi-)automatic content analysis of stances and frames in media texts

Digital Total — Oct. 2023, Hamburg, Germany
Few-shot learning for automated content analysis (FLACA)
in the German media debate on arms deliveries to Ukraine

DiTox'23 Workshop @LDK 2023 — Sept. 2023, Vienna, Austria
Debunking disinformation with GADMO: A topic modeling analysis
of a comprehensive corpus of German-language fact-checks

Statistische Woche 2023 — Sept. 2023, Dortmund, Germany
Bekämpfung von Desinformation durch GADMO: Analyse eines umfassenden
deutschsprachigen Faktencheck-Korpus mithilfe von Topic Modellen

Statistische Woche 2023 — Sept. 2023, Dortmund, Germany
Scrutinizing ChatGPT against few-shot learning with adapter extensions
and XLM-RoBERTa: A case study on identifying claims, arguments and their stance
in the German news media debate on arms deliveries to Ukraine

ECREA PolComm 2023 — Aug. 2023, Berlin, Germany
Beyond "Master Frames": A semi-automated approach to
studying viewpoint diversity of the media discourse

DGPuK 2023 May 2023
Few-shot learning for automated content analysis: Bremen, Germany
Efficient coding of arguments and claims in the debate on arms deliveries to Ukraine

MUFin'23 Workshop @AAAI 2023 Feb. 2023
Early warning systems? Building time consistent perception indicators Washington, DC, USA
for economic uncertainty and inflation using efficient dynamic modeling

SDP'22 Workshop @COLING 2022 Oct. 2022
Finding scientific topics in contionuously growing text corpora Gyeongju, Republic of Korea

Statistische Woche 2022 Sept. 2022
Monitoring consistent topics in continuously growing scientific text corpora Münster, Germany

Text2Story'22 Workshop @ECIR 2022 Apr. 2022
Dynamic change detection in topics based on rolling LDAs Stavanger, Norway

DAGStat 2022 Mar. 2022
Improving the reliability of LDA results using LDAPrototype as selection criterion Hamburg, Germany

EMNLP 2021 Nov. 2021
RollingLDA: An update algorithm of latent Dirichlet allocation to Punta Cana, Dominican Republic
construct consistent time series from textual data

EDML'20 Workshop @ECML PKDD 2020 Sept. 2020
Assessing the uncertainty of the text generating process using topic models Online

NLDB 2020 June 2020
Improving latent Dirichlet allocation: On reliability of the novel method LDAPrototype Online

Statistische Woche 2019 Sept. 2019
Quantifizierung der Stabilität der Latent Dirichlet Allocation Trier, Germany
mithilfe von Clustering auf wiederholten Durchläufen

DGPuK 2019 May 2019
Softwaretools für die Kommunikationsforschung Münster, Germany

DAGStat 2019 Mar. 2019
Measuring stability of replicated LDA runs Munich, Germany

Publications

Dissertation

- Rieger (2022). "Reliability evaluation and an update algorithm for the latent Dirichlet allocation". *TU Dortmund University*. DOI: 10.17877/DE290R-22949.

Peer-reviewed publications

- Schmidt, Rieger, Rahnenführer, Viciano, Wormer (2026). Beyond fact-checking: AI-supported quality assessment in journalism. Accepted for Studies in Communication and Media.
- Rieger, Muñoz, Grönberg, Lange, Ojeda-Pereira, Briceño, Nass, Stahl, Cassola, Rojas-Córdova, Keith-Norambuena, Lufin, Campos-Medina, Herrera-León, (2026). Following socio-environmental conflict narratives about energy transition in Chile: A spatio-temporal analysis using dynamic topic modeling. Accepted for Text2Story'26 Workshop.
- Rieger, Ruckdeschel, Wiedemann (2025). "PETapter: Leveraging PET-style classification heads for modular few-shot parameter-efficient fine-tuning". *Proceedings of the 21st KONVENS Conference*, pp. 333–348. URL: https://aclanthology.org/2025.konvens-1.29/.
- Rieger, Grönberg, Loschke, Braungardt (2025). "Monitoring narratives about the energy transition in Germany: Social (in)justice in the energy transition — from the digital debate to the living world". *Proceedings of the Text2Story'25 Workshop*. CEUR-WS. URL: https://ceur-ws.org/Vol-3964/.
- Müller, Blagov, Schmidt, Rieger, Jentsch (2025). "The macroeconomic impact of asymmetric uncertainty shocks". *The Journal of Economic Asymmetries* 31(e00410). DOI: 10.1016/j.jeca.2025.e00410.
- Loschke, Braungardt, Rieger (2025). "What motivates and demotivates energy savings in times of crisis? — An argument mining analysis using X/Twitter data". *Energy Efficiency* 18(4). DOI: 10.1007/s12053-024-10283-0.
- Rieger, Jentsch, Rahnenführer (2024). "LDAPrototype: A model selection algorithm to improve reliability of latent Dirichlet allocation". *PeerJ Computer Science* 10.2279. DOI: 10.7717/peerj-cs.2279.

- Lange, Rieger, Jentsch (2024). "Lex2Sent: A bagging approach to unsupervised sentiment analysis". *Proceedings of the 20th KONVENS Conference*, pp. 281–291. URL: https://aclanthology.org/2024.konvens-main.28/.
- Faymonville, Riffo, Rieger, Jentsch (2024). "spINAR: An R Package for Semiparametric and Parametric Estimation and Bootstrapping of Integer-Valued Autoregressive (INAR) Models". *Journal of Open Source Software* 9.97, p. 5386. DOI: 10.21105/joss.05386.
- Rieger, Yanchenko, Ruckdeschel, von Nordheim, Kleinen-von Königslöw, Wiedemann (2024). "Few-shot learning for automated content analysis: Efficient coding of arguments and claims in the debate on arms deliveries to Ukraine". *Studies in Communication and Media* 13, pp. 72–100. DOI: 10.5771/2192-4007-2024-1-72.
- Krause, Rieger, Flossdorf, Jentsch, Beck (2023). "Visually Analyzing Topic Change Points in Temporal Text Collections". *Vision, Modeling, and Visualization*. DOI: 10.2312/vmv.20231231.
- Rieger, Hornig, Flossdorf, Müller, Mündges, Jentsch, Elmer (2023). "Debunking Disinformation with GADMO: A Topic Modeling Analysis of a Comprehensive Corpus of German-language Fact-Checks". *Proceedings of the 4th Conference on Language, Data and Knowledge*, pp. 520–531. URL: https://aclanthology.org/2023.ldk-1.56.
- Rieger, Hornig, Schmidt, Müller (2023). "Early Warning Systems? Building Time Consistent Perception Indicators for Economic Uncertainty and Inflation Using Efficient Dynamic Modeling". *Proceedings of the 3rd Workshop on Modelling Uncertainty in the Financial World*. URL: https://github.com/JonasRieger/mufin23/blob/master/paper.pdf.
- Bittermann, Rieger (2022). "Finding scientific topics in continuously growing text corpora". *Proceedings of the 3rd Workshop on Scholarly Document Processing*, pp. 7–18. URL: https://aclanthology.org/2022.sdp-1.2.
- Lange, Rieger, Benner, Jentsch (2022). "Zeitenwenden: Detecting changes in the German political discourse". *Proceedings of the 2nd Workshop on Computational Linguistics for Political Text Analysis*. URL: https://old.gscl.org/en/arbeitskreise/cpss/cpss-2022/workshop-proceedings-2022.
- Rieger, Lange, Flossdorf, Jentsch (2022). "Dynamic change detection in topics based on rolling LDAs". *Proceedings of the Text2Story'22 Workshop*. CEUR-WS. URL: http://ceur-ws.org/Vol-3117/.
- Rieger, Jentsch, Rahnenführer (2021). "RollingLDA: An Update Algorithm of Latent Dirichlet Allocation to Construct Consistent Time Series from Textual Data". *Findings Proceedings of the 2021 EMNLP-Conference*. ACL, pp. 2337–2347. DOI: 10.18653/v1/2021.findings-emnlp.201.
- von Nordheim, Rieger, Kleinen-von Königslöw (2021). "From the Fringes to the Core - An Analysis of Right-Wing Populists' Linking Practices in Seven EU Parliaments and Switzerland". *Digital Journalism*, pp. 1–19. DOI: 10.1080/21670811.2021.1970602.
- von Nordheim, Koppers, Boczek, Rieger, Jentsch, Müller, Rahnenführer (2021). Die Entwicklung von Forschungssoftware als praktische Interdisziplinarität. *M&K Medien & Kommunikationswissenschaft* 69, pp. 80–96. DOI: 10.5771/1615-634X-2021-1-80.
- Rieger, Jentsch, Rahnenführer (2020). "Assessing the Uncertainty of the Text Generating Process Using Topic Models". *Proceedings of the ECML PKDD 2020 Workshops*. Vol. 1323. CCIS. Springer, pp. 385–396. DOI: 10.1007/978-3-030-65965-3_26.
- Rieger (2020). "ldaPrototype: A method in R to get a Prototype of multiple Latent Dirichlet Allocations". *Journal of Open Source Software* 5.51, p. 2181. DOI: 10.21105/joss.02181.
- Rieger, Rahnenführer, Jentsch (2020). "Improving Latent Dirichlet Allocation: On Reliability of the Novel Method LDAPrototype". *Natural Language Processing and Information Systems*, NLDB 2020. Vol. 12089. LNCS. Springer, pp. 118–125. DOI: 10.1007/978-3-030-51310-8_11.
- von Nordheim, Rieger (2020). "Im Zerrspiegel des Populismus - Eine computergestützte Analyse der Verlinkungspraxis von Bundestagsabgeordneten auf Twitter". *Publizistik* 65, pp. 403–424. DOI: 10.1007/s11616-020-00591-7.

Selected non-peer reviewed publications (preprints, working papers and datasets)

- Shrub, Rieger, Müller, Jentsch (2022). "Text data rule - don't they? A study on the (additional) information of Handelsblatt data for nowcasting German GDP in comparison to established economic indicators". *Ruhr Economic Papers* #964. DOI: 10.4419/96973128.
- Jentsch, Mammen, Müller, Rieger, Schötz (2021). "Text mining methods for measuring the coherence of party manifestos for the German federal elections from 1990 to 2021". *DoCMA Working Paper* #8. DOI: 10.17877/de290r-22363.
- Rieger, von Nordheim (2021). "corona100d - German-language Twitter dataset of the first 100 days after Chancellor Merkel addressed the coronavirus outbreak on TV". *DoCMA Working Paper* #4. DOI: 10.17877/DE290R-21911.

Teaching Experience (at TU Dortmund University, if not specified)

Natural Language Processing | *Lecture (English)* — WiSe 2023/24 – 2025/26
Data Mining Cup | *Seminar (English)* — SuSe 2019 – 2025
Introduction to Topic Modeling | *Seminar (English)* — SuSe 2023
University of Bremen

Text as Data | *Lecture (English)* — WiSe 2022/23
Einführung in LaTeX | *Compact course (German)* — SuSe 2019 – 2022
Fallstudien I | *Seminar (German)* — WiSe 2021/22
Schätzen und Testen | *Organization (German)* — WiSe 2021/22
Nichtparametrische Verfahren | *Exercise (German)* — WiSe 2020/21
Text Data meets Econometrics | *Seminar (English)* — WiSe 2020/21
Entscheidungstheorie - Statistik VI | *Exercise (German)* — SuSe 2020
Wahrscheinlichkeitstheorie - Statistik V | *Exercise (German)* — WiSe 2019/20
Textdatenanalyse | *Seminar (German)* — SuSe 2019

Supervised Theses

Master

- (Nishat Tasnim Ahmed Meem, planned)
- (Ahmet Emircan Coşkun, planned)
- Comparing denoising and masking strategies in self-supervised time series forecasting with transformers (Yat Chun Fung, running) — *2nd Reviewer*
- Comparative analysis of products perception using unsupervised machine learning and LLMs (Vipul Chauhan, 2025) — *1st Reviewer*
- Stock recommendation using graph neural networks and economy news data (Lars Grönberg, 2025) — *Advisor*
- Data preparation & training of LLMs in e-commerce settings (Mariia Hrechyn, 2024) — *1st Reviewer*
- Exploration of unsupervised language-style transfer (ULST) methods (Rohan Kumar Nayak, 2024) — *1st Reviewer*
- Die Zukunft des Sportjournalismus: Einsatz von Sprachmodellen zur Erstellung von Spielberichten (Niklas Herzog, 2024) — *1st Reviewer*
- ChatGPT as a negotiator: Analyzing its adherence to principles of proportionality and equality (Veronika Tsishetska, 2024) — *1st Reviewer*
- Attaching PET (-like) models to RELATIO to find causal relationships between narratives (Muhammad Mahir Hasan Chowdhury, 2024) — *1st Reviewer*
- Semantic shift modelling with graph neural networks (Imene Kolli, 2024) — *2nd Reviewer*
- Diachronic sense modeling with hierarchical word embeddings (Aymane Hachcham, 2024) — *2nd Reviewer*
- #FrierenFürDenFrieden: Quantifizierung des Diskurses zum Thema Gas- und Energiesparen auf Twitter durch Anwendung von NLP-Methoden und Textklassifizierung durch RoBERTa (Carmen Loschke, 2023) — *Advisor*
- Comparison of diachronic embeddings with pre-trained model embeddings for historical texts (Priyanka Madiraju, 2023) — *1st Reviewer*
- Text Data-based nowcasting of German GDP growth using newspaper data (Yuliya Shrub, 2022) — *Advisor*
- Resampling strategies for unsupervised sentiment analysis using lexicon-based text embedding methods (Kai-Robin Lange, 2021) — *Advisor*

Bachelor

- Lexicon-based text clustering using document embeddings (Jan Plotzki, running) — *2nd Reviewer*
- Können Zeitungsartikel in ökonometrischen Vorhersagen helfen? Eine VAR-basierte Prognose wirtschaftlicher Kennzahlen mithilfe der IPI- und UPI-Indizes (Kjell Noack, running) — *1st Reviewer*
- Modellierung der Zielgruppe und Beliebtheit in Deutschland veröffentlichter Manga — Vergleich von Modellen auch mithilfe von Textdatenanalyse (Darya Lukashina, 2024) — *1st Reviewer*
- Einsatz von Adversarial-Examples zur Untersuchung verschiedener Machine-Learning-Algorithmen zur Bewertung von Divergent Thinking Tasks (Yannick Hilker, 2023) — *2nd Reviewer*
- Modellierung von Häufigkeiten medizinischer Themen in wissenschaftlichen Veröffentlichungen am Beispiel Brustkrebs (Greta Schlautmann, 2023) — *2nd Reviewer*
- Comparison of active learning techniques for the benefit of data set generation in the field of text mining (Jannik Bloß, 2023) — *1st Reviewer*